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SUPERSEDING
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MILITARY SPECIFICATION

CHASSIS, TRAILER 2-WHEEL CART TYPE 1/4 TO 3 1/2 TON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers 2-wheel trailer chassis intended for mounting of various type bodies and equipment used by the United States Armed Forces.

1.2 Classification. Chassis shall be of the following models as specified (see 6.2):

Model	Description
M569A1	1/4 Ton
M116A2	3/4 Ton
M514	1 Ton
XM545	1 Ton
M103A3	1 1/2 Ton
M103A4	1 1/2 Ton
M492A1	1 1/2 Ton

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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Model	Description
XM580	1 1/2 Ton
M390	2 Ton
M390C	2 Ton
M200A1	2 1/2 Ton
M537	2 1/2 Ton
M353	3 1/2 Ton
M536	3 1/2 Ton

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation

SPECIFICATIONS
FEDERAL

VV-L-800	- Lubricating Oil, General Purpose, Preservative (Water Displacing, Low Temperature).

MILITARY

MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-L-3150	- Lubricating Oil, Preservative, Medium.
MIL-H-3992	- Hose and Hose Assembly Rubber: Air and Vacuum Brake, Automotive.
MIL-H-5606	- Hydraulic Fluid, Petroleum Base, Aircraft, Missile, and Ordnance.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-H-13719	- Hose Assembly, Rubber: Hydraulic Brake.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-C-46164	- Rustproofing for Military Vehicles and Trailers.
MIL-B-46176	- Brake Fluid, Silicone Automotive All Weather, Operational and Preservative.
MIL-V-62038	- Vehicle, Wheeled: Preparation for Shipment and Storage of.

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STANDARDS
FEDERAL

FED-STD-595	- Colors.

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MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-642	- Identification Marking of Combat and Tactical Transport Vehicles.
MIL-STD-721	- Definitions of Terms for Reliability and Maintainability.
MIL-STD-882	- System Safety Program Requirements.
MIL-STD-1179	- Lamps, Reflectors and Associated Signaling Equipment for Military Vehicles.
MIL-STD-45662	- Calibration Systems Requirements.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise speciifed, the issues shall be those in effect on the date of the solicitation.

DRAWINGS
ARMY

8358991	- Chassis, Trailer, 1 1/2 Ton, 2-Wheel, M103A3.
8358992	- Chassis, Trailer, 1 1/2 Ton, 2-Wheel, M103A4.
8358998	- Chassis, Trailer, 2 1/2 Ton, 2-Wheel, M200A1.
8736243	- Chassis, Trailer, 1 Ton, 2-Wheel, M514.
8736268	- Chassis, Trailer, 3 1/2 Ton, 2-Wheel, M353.
8736315	- Chassis, Trailer, 2 Ton, 2-Wheel, M390C.
8736321	- Chassis, Trailer, 3 1/2 Ton, 2-Wheel, M536.
8736322	- Chassis, Trailer, 2 1/2 Ton, 2-Wheel, M537.
8736324	- Chassis, Trailer, 2 Ton, 2-Wheel, M390.
8736332	- Chassis, Trailer, 1 Ton, 2-Wheel, XM545.
8736417	- Chassis, Trailer, 1 1/2 Ton, 2-Wheel, XM580.
8736996	- Chassis, Trailer, 1/4 Ton, 2-Wheel, M569A1.
8750054	- Chassis, Trailer, 3/4 Ton, 2-Wheel, M116A2.
8736829	- Chassis, Trailer, 1 1/2 Ton, 2-Wheel, M492A1.

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

DEPARTMENT OF TRANSPORTATION (DOT)

393.70(d) -Federal Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Department of Transportation, Federal Highway Administration, Washington, DC 20591).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish the first completed chassis which shall be subjected to first article inspection (see 4.3). First article inspection samples, properly marked with identifying information shall be representive of the units to be furnished to the Government. All subsequent chassis delivered to the Government shall conform to this sample in all their pertinent physical and performance attributes.

3.2 Materials. Materials shall be as specified herein and in referenced specifications and drawings. Materials shall be free of defects and imperfections that might affect the serviceability and appearance of the finished product (see 4.8.1, 4.8.2 and 6.3).

3.2.1 Qualified products. When a part or component is specified to conform to a specification having a Qualified Products List (QPL), the contractor shall maintain documentation of item acquisition from such QPL. The documentation shall include the QPL date and identification of the supplier, purchase order, and quantity (see 4.8.1).

3.2.2 Ozone resistance. When rubber components are required to be ozone resistant, the degree of resistance shall be as specified in applicable specifications, or drawings (see 4.8.1).

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3.3 Construction. Chassis components, chassis sub-assemblies and assemblies shall be fabricated and assembled into a complete unit in accordance with Army drawings specified below (see 4.8.1 and 4.8.2):

Drawing	Model
8358991	- M103A3.
8358992	- M103A4.
8358998	- M200A1.
8736243	- M514.
8736268	- M353.
8736315	- M390C.
8736321	- M536.
8736322	- M537.
8736324	- M390.
8736332	- XM545.
8736417	- XM580.
8736996	- M569A1.
8750054	- M116A2.
8736829	- M492A1.

3.3.1 Bearing seals. When fording or operating in mud, sand and snow, seals shall prevent entrance of foreign matter into the bearings, which are exposed to contamination during these operations. All bearing seals shall restrict the leaking of lubricants from bearings (see 4.8.2 and 4.8.2.1).

3.3.2 Electrical system.

3.3.2.1 Electrical circuits. Electrical circuits shall maintain continuity from end to end without evidence of internal or external shorts during all operating conditions (see 4.8.2).

3.3.2.2 Lights. All lights specified in applicable drawings shall operate throughout all operating conditions and shall meet the requirements of MIL-STD-1179 (see 4.8.2).

3.3.3 Controls. All electrical, mechanical, and hydraulic controls shall operate without malfunction, throughout all ranges of operation, under all operating conditions (see 4.8.2).

3.3.4 Adjustment mechanisms. All adjustment mechanisms shall function properly and maintain adjustment settings during all chassis operating conditions (see 4.8.2).

3.3.5 Welding and welding repairs. Welding and welding repairs of any type shall be made only when, and to the extent, specifically authorized, by the procuring activity (see 4.3.1.1, 4.8.1, and 4.8.2).

3.3.6 Safety. No condition shall exist which may create a safety hazard to operating or maintenance personnel (see 4.8.2 and 4.8.2.2).

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3.3.6.1 Safety chains. The trailer shall be equipped with two safety chains conforming to Federal Motor Carrier Safety Regulations 393.70(d) (see 4.8.1, 4.8.2 and 4.8.2.2.1).

3.3.7 Servicing and adjustment. Prior to acceptance by the Government, the contractor shall adjust and service each chassis for immediate operational use, including at least the following: adjust braking system, check electrical system, inflate all tires and completely lubricate the trailer and all running gear with grades of lubricants recommended in table I for the ambient temperature at the delivery point (see 4.8.1, 4.8.2 and 4.8.2.3).

TABLE I. Special products specifications.

Product use	Ambient air temperatures	
	-65° to 0°	-10° to +132°F
Oil:		
For hydraulic brakes	MIL-B-46176	MIL-B-46176
For shock absorbers	MIL-J-5606	MIL-H-5606
For general purpose lubrication	VV-L-800	MIL-L-3150
Grease:		
For sealed bearings	MIL-G-23827	MIL-G-23827
For general chassis lubrication, including wheel bearings	MIL-G-10924	MIL-G-10924

3.3.8. Service brakes. When specified (see 6.2), the chassis trailer shall be equipped with service brakes. Brake hose and brake hose assemblies shall meet the requirements of MIL-H-3992 and MIL-H-13719, as applicable (see 4.8.1 and 4.8.2).

3.3.9 Landing device. Unless otherwise specified (see 6.2), the chassis shall be equipped with swing-up type landing device. The vertically adjustable swing-up type landing device shall be furnished and mounted on the M103A3, M353, M390, M390C, M200A1, M492, M536, and XM580 chassis (see 4.8.1 and 4.8.2).

3.4 Performance. Unless otherwise specified herein, chassis shall meet performance requirements of this specification when equipped as specified herein, serviced with standard products specified in table I, loaded with rated payload and coupled to the tactical prime mover specified in table II. Chassis, serviced and equipped for existing climatic conditions, shall operate as specified without special equipment (see 4.8.3).

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TABLE II. Payload and speeds.

Chassis	Preferred Companion prime mover	Payload (pounds)	Miles per Hour		
			Highway	Gravel roads	Cross-country
M569A1 1/	M151	500	50	35	15
M514	M536	2500	50	-	10
M116A2	1 1/4 ton truck	1965	50	35	15
XM545	M36	2800	50	35	10
M103A3	M35	4090	50	35	10
M103A4	M35	4090	50	35	10
M492A1	M35	4200	50	-	15
M390	M35	4000	50	35	10
M390C	M35	4000	50	35	10
M200A1	M35	5000	50	35	15
M353	M54	7000	50	35	15
M536	M52	6800	50	35	10
M537	M35	5000	50	35	10
XM580	M35	4000	50	35	10

1/ The durability test shall be conducted with the mission payload, however, if such payload is not available, the loads cited herein shall be applied.

3.4.1 Maintainability. A measure of maintainability parameters should be in accordance with the definition in MIL-STD-721. The total active maintenance (scheduled and unscheduled) excluding driver/crew checks shall not exceed 16 man-hours during the first 6000 miles when operated in accordance with table II. This equates to a Maintenance Ratio (MR) of 0.0026 man-hours per operating hours (see 4.8.3.1).

3.4.2 Durability.

3.4.2.1 Chassis with rated payload. Except as specified in 3.4.4.2, chassis with rated payload shall possess not less than 0.60 probability of completing a minimum of 6000 miles of operation without experiencing a failure of principal components or subassemblies. This shall include, suspension (including springs, axles and wheels), brake systems, landing device, leveling jacks (when specified), lunette and frame. Operation shall be conducted in accordance with parameters of table II and shall be apportioned as per table III (see 4.8.3.2).

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TABLE III. 6000 mile test, loaded as specified in table II

Course	Mileage and speeds	Chassis payload
Hard-surfaced roads	2400 miles at varying speeds up to maximum	Highway
Gravel and dirt roads	1500 miles at varying speeds up to maximum	Highway
Level cross-country	1380 miles at varying speeds up to maximum	Cross-country
Hilly cross-country	600 miles at varying speeds up to maximum	Cross-country
Belgian block	120 miles at speeds applicable to chassis characteristics	Cross-country

3.4.2.2 Chassis without payload. (see 3.4.4.2).

3.4.2.3 Malfunction. Any malfunction which the operator or crew (user) can remedy will not be considered a failure provided that the repair is authorized or prescribed as an operator function and can be accomplished within 15 minutes using only controls, tools, or spare parts incorporated in, or carried with, the end item or its prime mover.

3.4.2.4 Failure definition. A failure is defined as any malfunction of an end item that requires corrective action which cannot be deferred for the remainder of its 6000 miles, if the malfunction is of such consequence as to require overhaul, replacement, rebuild, or salvage as prescribed by maintenance manual.

3.4.2.5 Corrective action. Corrective action is not deferrable if the malfunction causes (or would cause if not corrected):

a. Inability to commence operation, cessation of operation, or inability to perform in compliance with vehicle specification requirements.

b. A critical or catastrophic hazard as defined by MIL-STD-882.

3.4.2.6 Maintenance chargeability. Chargeable active scheduled maintenance includes both servicing time and inspection time above crew level or the estimated man-hours that should have been performed at every scheduled maintenance interval. This includes the scheduled maintenance man-hours required or estimated at the end of the test in bringing the vehicle to restricted use of new, used, repaired, or reconditioned material that is serviceable and issuable for its intended purpose. Chargeable active unscheduled maintenance includes diagnostic time, fault correction time, adjustment/calibration time, and check out/inspection time, for on-vehicle repairs only. This maintenance must be the result of hardware or technical

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manual problems and will not include errors, checks, services or repairs by the crew, maintenance errors or modifications, test time abuse or man-hours spent in engineering investigations or directed actions to verify a particular problem. It excluded the man-hours spent during initial inspection and final inspection if the malfunction is discovered after satisfactory completion of the test mileage. However, if the malfunction is discovered before the completion of the test which requires corrective action but postponed just to complete the test, the estimated or required maintenance man-hours are considered to be part of chargeable unscheduled maintenance (see 4.8.3.2).

3.4.3 Environmental. The chassis shall operate under extreme conditions of weather and altitude in ambient air temperatures ranging from minus 50 to plus 125 degrees Fahrenheit (°F). The complete chassis, when in storage, shall withstand a temperature of minus 80°F without deterioration that may cause failure of any component part of the chassis (see 4.8.3.3).

3.4.4 Payload, towing speeds, and trailing ability.

3.4.4.1 Highway (paved) and gravel roads, payload and speeds. The chassis shall be loaded with the highway rated payload, or with a simulated load of equal weight, uniformly distributed over load area. Chassis shall be towed over relatively level, smooth, improved, and prepared hard-surfaced roads and gravel roads at average speeds specified in table II, without damage to chassis or prime mover. When loaded and towed at speeds specified, in a straight line, on a level, smooth, paved surface, the path of the towed chassis shall not deviate more than 3 inches to either side of the path of the towing vehicle. When the payloads for trailers M514, XM545, M390, M390C, M536 and M537 are simulated, the loads shall be uniformly distributed to result in 450 to 550 pounds of static weight over the lunette (see 4.8.3.4).

3.4.4.2 Operation without payload. The 6000 mile durability test shall include 50 miles of gravel and 30 miles of Belgian block without payload, at start and completion of foregoing operations, for a total of 160 miles (see 4.8.3.4).

3.4.4.3 Payload and speeds. The chassis loaded with rated payload uniformly distributed over the load area, shall be capable of being towed over unimproved roads, trails, open fields, rolling hills, and cross-country terrain at sustained speeds specified in table II, without damage to chassis or the prime mover. The payload shall be distributed so that the proper center of gravity is established (see 4.8.3.5).

3.4.5 Grade and slope operation. The chassis shall operate without slipping or upsetting when ascending or descending longitudinal grades up to 40 percent and side slopes up to 20 percent (see 4.8.3.6).

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3.4.6 Brake holding ability.

3.4.6.1 Service brakes. The combined service brakes of the chassis and towing vehicle shall, under all conditions of loading, bring the combination to a stop within a distance of 30 feet from a speed of 20 miles per hour (mph) on dry, smooth, level road, free of loose material (see 4.8.2 and 4.8.3.7).

3.4.6.1.1 Automatic brake actuation. An automatically actuated device shall apply the chassis brakes upon breakaway from the towing vehicle, and shall maintain application of brakes for not less than 15 minutes (see 4.8.3.8).

3.4.6.2 Parking brake. The parking (hand) brakes shall be capable of holding the chassis on a dry, hard surfaced, 30 percent grade with payload headed up or down grade (see 4.8.2 and 4.8.3.9).

3.4.7 Landing device. The landing device on all chassis shall have adequate capacity to support the fully loaded vehicle on any grade up to 10 degrees in any direction; and shall withstand without damage the strain imposed upon it when coupling or uncoupling from the towing vehicle. The landing device shall extend and retract; and when retracted shall provide ground clearance for cross-country operation (see 4.8.2 and 4.8.3.10).

3.4.8 Leveling jacks. When specified (see 6.2), the manually operated leveling jacks shall have capacity to raise and support the fully loaded chassis and shall adjust to permit placing the jack bases on the ground and leveling the chassis on an off-the-road site (see 4.8.2 and 4.8.3.10).

3.4.9 Turning and tracking ability. The chassis, when coupled to the prime mover operating in its minimum turning circle, shall follow without cramping and without damage to the towed chassis or prime mover and without interference to the towed chassis or prime mover (see 4.8.3.11).

3.4.10 Fording ability. The chassis shall be towable through fresh or salt water to a depth great enough to entirely submerge the chassis for periods up to 15 minutes duration without damage to the chassis or its components. This shall be accomplished without the use of special kits and with a minimum of preparation and servicing before and after fording operations (see 4.8.3.12).

3.4.11 Stabilizers. Stabilizers such as step jacks on M200A1 and rear support legs (props) on M103A3 and M116A2, shall stabilize the fully loaded chassis, without raising chassis wheels off the ground. The stabilizers shall exhibit adjustment to permit placing their bases to the ground for stabilizing the chassis on an off-the-road site (see 4.8.2 and 4.8.3.13).

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3.5 Painting and rustproofing.

3.5.1 Painting. Unless otherwise specified (see 6.2), chassis and component assemblies and parts which require painting shall be prepared and painted with a chemical agent resistant coating (CARC). The materials used for specific service shall match the specified color chip of FED-STD-595. When specified (see 6.2), camouflage pattern shall conform to the pattern stipulated (see 4.8.1 and 4.8.2).

3.5.2 Rustproofing. Unless otherwise specified (see 6.2), the chassis shall be rustproofed in accordance with MIL-C-46164 (see 4.8.1 and 4.8.2).

3.6 Identification plates and marking.

3.6.1 Identification plates. Identification plates shall meet the requirements of MIL-P-514. Any special information required by the using service shall be as specified by the procuring activity (see 4.8.2 and 6.2).

3.6.2 Marking. Chassis marking shall be in accordance with MIL-STD-642. All parts, assemblies, and subassemblies requiring identification shall be identified in accordance with MIL-STD-130 (see 4.8.2).

3.7 Workmanship. Workmanship shall be of such quality as to assure that the chassis and its components are free of defects that compromise, limit or reduce the capability of the chassis in the performance of its intended use (see 4.8.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the

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specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10 percent of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

4.1.3 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operation to determine that practices, methods, and procedures of the written quality assurance plan are being properly applied, and (b) Government product inspection to measure the quality of product offered for acceptance. Deviation from the prescribed or agreed upon procedures, or instances of poor practices which might have an adverse effect upon the quality of the product, will immediately be called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies shall be cause for suspension of acceptance until corrective action has been made, or until conformance of product to prescribed criteria has been demonstrated.

4.1.4 Quality assurance provisions. In the conduct of inspection, the contractor shall adhere to Quality Assurance Provisions (QAP) and General Quality Assurance Provision (STA Form 458) as applicable and as required by the documents forming part of this specification.

4.1.4.1 Supplementary quality assurance provision (SQAP). Drawing numbers specified in 3.3, forming part of this specification, designate additional "Master List of Supplementary Quality Assurance Provision (SQAP)" with the same Army drawing number. In the conduct of inspection, the contractor shall adhere to SQAP provisions as applicable and required by those drawings as specified herein.

4.1.5 Certification. Where certification (see 4.8.1) is required to verify material or component conformance to a specification, the contractor shall maintain such certification along with documented test results and performance and analytical data, as applicable.

4.2 Classification of inspection:

a. First article inspection (see 4.3).
   1. First production chassis inspection (FPCI) (see 4.3.1).
   2. Initial production test (IPT) (see 4.3.2).

b. Quality conformance inspections (QCI) (see 4.4).
c. Control tests (see 4.5).
d. Comparison tests (see 4.6).

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4.3 First article inspection. First article inspection shall be performed on the first production chassis and on 2 additional chassis randomly selected from the first month's production or from the first 20 produced, which shall be designated as initial production chassis. Approval of the first article samples shall not relieve the contractor of the obligation to supply chassis that are fully representative of those inspected as a first article sample. Any change or deviation of production vehicles from the first article sample shall be subject to the approval of the contracting officer.

4.3.1 FPCI.

4.3.1.1 In-process inspection. The first production chassis shall be inspected during fabrication to determine conformance of materials and workmanship to specified requirements. These inspections shall preceed application of primer and paint. Processing and welding procedures, the quality system, inspection records, calibration procedures, radiographic procedures and welder certifications shall be reviewed and evaluated.

4.3.1.2 Completed chassis inspections.

4.3.1.2.1 Contractor inspection. The first production chassis on each contract shall be road tested and completely inspected by the contractor for conformance to the requirements specified herein. The applicable road test shall be that specified in 4.4.4. Upon completion, the contractor, shall submit the chassis (and make available all inspection records and certifications) to the responsible Government inspection element at contractor's plant for preliminary inspection.

4.3.1.2.2 Preliminary inspection. The responsible Government inspection element will conduct the FPCI specified in table IV to the extent that acceptability of the chassis is verified.

4.3.1.2.3 Provisional inspection. The Government procuring activity will perform the FPCI specified in table IV to the degree required to justify approval of the chassis.

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TABLE IV. Classification of inspections.

Title	Requirements	Inspection	First article		1/ QCI	1/ Control	2/ Comparison
			1/ FPCI	2/ IPT			
Materials and construction	3.2 thru 3.3, 3.3.5, 3.3.6.1, 3.3.7, 3.3.8, 3.3.9, 3.5.1, 3.5.2	4.8.1	X				
Defects (see table V)	3.2, 3.3, 3.3.1 thru 3.3.9, 3.4.6.1, 3.4.6.2, 3.4.7, 3.4.8, 3.4.11, 3.5.1, 3.5.2, 3.6, 3.7	4.8.2	X	X	X		
Bearing seals	3.3.1	4.8.2.1	X	X			X
Safety	3.3.6	4.8.2.2	X	X			X
Safety chains	3.3.6.1	4.8.2.2.1	X	X			X
Servicing and adjustment	3.3.7	4.8.2.3	X	X			X
Maintainability	3.4.1	4.8.3.1		X			
Durability	3.4.2	4.8.3.2		X			
Environmental	3.4.3	4.8.3.3		X			X
Highway payload and speeds	3.4.4.1	4.8.3.4		X			X
Operation without payload	3.4.4.2	4.8.3.4		X			X
Cross-country payload and speeds	3.4.4.3	4.8.3.5		X			X
Grade and slope operation	3.4.5	4.8.3.6		X			X
Service brakes	3.4.6.1	4.8.3.7	X	X		X	X
Automatic brake actuation	3.4.6.1.1	4.8.3.8	X	X		X	X
Parking brake	3.4.6.2	4.8.3.9	X	X		X	X
Landing device	3.4.7	4.8.3.10	X	X		X	X
Leveling jacks	3.4.8	4.8.3.10	X	X			
Turning and tracking ability	3.4.9	4.8.3.11	X	X			X
Fording ability	3.4.10	4.8.3.12		X			X

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TABLE IV. Classification of inspections - Continued.

Title	Requirements	Inspection	First article 1/ FPCI	First article 2/ IPT	1/ QCI	1/ Control	2/ Comparison
Stabilizers	3.4.11	4.8.3.13	X	X		X	X
Treatment and painting	3.5.1	4.8.2	X	X		X	X
Rustproofing	3.5.2	4.8.2	X	X		X	X
Identification plates	3.6.1	4.8.2	X	X		X	X
Marking	3.6.2	4.8.2	X	X		X	X
Workmanship	3.7	4.8.2	X	X		X	X
Packaging	Section 5	4.8.4		X			X

1/ At place of manufacture
2/ At Government proving ground

4.3.1.2.4 Chassis disposition. After completion of inspection, the first production chassis shall remain at the manufacturing facility, as a production sample, and be the last chassis of its type shipped on a contract. The chassis may be released sooner at the discretion of the contracting officer. The contractor shall service and maintain the chassis during this period in accordance with applicable documents for care and preservation while in storage. All configuration changes taking place after the first production inspection, shall be made to the first production chassis so that the chassis will be representative of the current configuration throughout the life of the contract. No configuration changes may be implemented on production vehicles after Government conditional acceptance of the first production chassis (applicable chassis type) without written authorization from the Government.

4.3.2 IPT.

4.3.2.1 Contractor inspection. The contractor shall inspect the initial production test chassis in accordance with 4.4.

4.3.2.2 Government inspection. The Government shall perform the IPT specified in table IV and a 6000 mile test as specified in table III loaded as specified in table II.

4.3.3 Deficiencies. Unresolved failures or deficiencies during, or as a result of, first article inspection shall be cause for rejection of the vehicles until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of the IPT shall be evidence that all vehicles already accepted prior to completion of the IPT are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all vehicles shall be corrected by the contractor.

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4.4 Quality conformance inspection (QCI). QCI shall be performed in the following sequence and utilizing an approved Final Inspection Record (FIR).

a. Examination for the defects specified in table IV and classified in table V.
b. A 50-mile road test in accordance with 4.4.4.
c. QCI specified in table IV.

4.4.1 One hundred percent. Each vehicle shall be subjected to the QCI specified in 4.4.

4.4.2 Acceptable quality level. When authorized by the Government contracting authority, one hundred percent QCI (see 4.4.2.1) shall be waived. Thereafter, samples shall be selected in accordance with 4.4.2.1.2 and shall be inspected to determine conformance to the following acceptable quality levels (AQL).

Classification	AQL
Major	25/100 units
Minor	150/100 units

TABLE V. Classification of defects.

Categories	Defects	Method of inspection
Critical	None defined	
Major	AQL 25/100	
101	Dimensions affecting interchangeability, out of tolerance (see 3.2).	SIE 1/
102	Assembly incomplete (see 3.3).	Visual
103	Malfunction; inoperative; lighting sequence improper, damage to electrical system components (see 3.3.2.1 and 3.3.2.2).	Visual and functional
104	Structual, welding and riveting defects of frame (see 3.3.5).	Visual
105	Incorrect lengths; components damage to safety chains, intervehicular hose, cables, tubing wiring, and ground straps (see 3.3.6 and 3.3.6.1).	Visual and functional
106	Malfunction; component damage; nonconformance to holding of service and parking brakes (see 3.3.8, 3.4.6.1 and 3.4.6.2).	Visual and functional
107	Malfunction, nonconformance of automatic actuation (brake) (see 3.4.6.1).	Functional
108	Improper locking action (raised or lowered) of landing device (see 3.4.7).	Visual and functional

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TABLE V. Classification of defects - Continued.

Categories	Defects	Method of inspection
109	Malfunction of extended, retracted, or stowed leveling jacks (see 3.4.8).	Functional
110	Malfunction; component damage; incorrect assembly of stabilizing legs (see 3.4.11).	Visual and functional
111	Damage to suspension system.	Visual
112	Improper assembly or installation of axles.	Visual and functional
113	Tire, wheel damage.	Visual
114	Casting defects, size of lunette eye.	Visual and gage
115	Overheating of operating units.	Tactile (touch)
116	Identification marking improper (see 3.6).	Visual
117	Faulty workmanship affecting performance	Visual
Minor	AQL 150/100	
201	Dimensions not affecting interchangeability, out of tolerance (see 3.2).	SIE 1/
202	Incorrect type, size or mounting of tires and wheels.	Visual
203	Improperly coded, protected, installed or assembled electrical cables, tubing, wires, and ground straps.	Visual
204	Incorrect assembly, installation or clearance of brakes, system components.	Visual
205	Incorrect assembly or installation of electrical system components.	Visual
206	Incorrect assembly or installation of suspension system components.	Visual
207	Incorrect assembly or installation of frame and sheet metal.	Visual
208	Incorrect assembly or installation of landing device.	Visual
209	Incorrect assembly or installation of lunette eye.	Visual
210	Missing, incorrectly located; welding defects of tiedowns, lashing hooks, and lifting devices.	Visual
211	Missing, defective, incorrect installation of reflectors.	Visual
212	Incorrect application; color of paint.	Visual
213	Defective; missing, improperly installed lube fittings.	Visual
214	Incomplete data; improper size or installation of decals, data and instruction plates (see 3.6.1 and 3.6.2).	Visual
215	Improper application of lubrication.	Visual

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TABLE V. Classification of defects - Continued.

Categories	Defects	Method of inspection
216	Incorrect level and grade of hydraulic fluid.	Visual
217	Incorrect application, coverage and materials for protective coating.	Visual
218	Missing, improperly processed records and forms.	Visual
219	Missing, improperly processed tools.	Visual
220	Faulty workmanship affecting appearance	Visual

1/ SIE Standard inspection equipment.

4.4.2.1 Sampling.

4.4.2.1.1 Lot formation. An inspection lot shall consist of all the chassis produced in an identifiable production period (one day) submitted at one time for acceptance.

4.4.2.1.2 Sampling for inspection. When sampling is authorized for QCI (see 4.4.2) samples shall be selected in accordance with general inspection level II and MIL-STD-105. Before sampling may be initiated, the contractor shall examine at least 20 consecutively produced chassis. The examination shall establish that the process average percent defective, as defined in MIL-STD-105, is not greater than the specified AQL.

4.4.3 Defects. The following rules shall be applied to the counting of defects.

4.4.3.1 Unclassified defects. All defects that have no effect on function, safety, interchangeability or life, that are considered departures from good workmanship will be noted in writing. Workmanship deficiencies falling within this category and recurring in five consecutive lots, or ten lots or more within a thirty-day period, will be added to the minor defects classification with no increase in AQL's.

4.4.3.2 Recurring major deficiencies. A major deficiency is recurring when the same defects occurs more than once in the same sample, or when the defect occurs in two successive samples. A major defect may be considered recurring when the historical inspection records ("P" chart or approved equivalent) reflect such a condition. Recurring major deficiencies shall be cause for the entire lot, or lots, to be inspected for the recurring defieicncies. The deficiencies shall be corrected by the contractor when found.

4.4.3.3 Recurring minor deficiencies. A minor deficiency is recurring if it occurs more than twice in the same sample, or when the defect occurs in four successive samples. Recurring minor deficiencies shall be cause for the entire lot or lots to be inspected to the recurring deficiencies and correction shall be accomplished prior to acceptance.

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4.4.4 Road test. Chassis shall be loaded with full or simulated payload, operated for a distance of 50 miles, and subjected to all control tests specified in table IV. These tests shall be performed at the place of manufacture by the contractor. The test course shall be a relatively smooth, level, hard-surfaced road.

4.5 Control tests. Control tests shall be conducted on one chassis per month from each lot of 100 chassis consecutively produced, except that not more than 2 tests shall be performed in a 30-day period. The chassis shall be road tested in accordance with 4.4.4. The chassis shall subsequently be subjected to the control tests specified in table IV.

4.6 Comparison test. The Government may randomly select a chassis at any time during the contract production period and subject these chassis to the comparison tests specified in table IV as well as 6000 mile test in accordance with 4.3.2.2. Tests will be conducted by the Government at a site it will select. Vehicles selected for comparison tests shall not have been previously tested according to 4.5, but shall have passed inspection according to 4.4.

4.7 Failure. Failure of any vehicle to pass any of the specified inspections or failure to meet inspection AQL levels, as applicable, shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.8 Methods of inspection.

4.8.1 Materials and construction. Conformance to 3.2 thru 3.3, 3.3.5, 3.3.6.1, 3.3.7, 3.3.8, 3.3.9, 3.5.1, and 3.5.2 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.8.2 Defects. Conformance to 3.3.1 through 3.3.9, 3.4.6.1, 3.4.11, 3.5.1, 3.5.2, 3.6 and 3.7 shall be determined by examination for the defects listed in table V. Examination shall be visual, tactile, or by measurement with standard inspection equipment.

4.8.2.1 Bearing seals. To determine conformance to 3.3.1, all bearings, as applicable, shall be examined for lubricant leakage and for contamination of lubricant.

4.8.2.2 Safety. To determine conformance to 3.3.6, all exposed parts which are energized electrically shall be located, insulated, fully enclosed or guarded so as to prevent hazards to operating personnel and equipment functioning. All moving parts which are of such nature, or so located, as to be a hazard to operating or maintenance personnel shall be enclosed or guarded. Protective devices shall not impair operating functions.

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4.8.2.2.1 Safety chains. To determine conformance to 3.3.6.1, all safety chains shall be examined for conformance to requirements.

4.8.2.3 Servicing and adjustment. To determine conformance to 3.3.7, servicing and adjustment of specified systems shall be examined.

4.8.3 Performance. Conditions for performance shall be as specified in 3.4.

4.8.3.1 Maintainability. To determine conformance to paragraph 3.4.1, the Maintenance Ratio of the test chassis shall not exceed 16 man-hours during the first 6000 miles of the IPT. Maintenance operations will be performed at the Government approved test site as per technical manual instructions.

4.8.3.2 Durability. To determine conformance to 3.4.2, the durability requirement will be verified at a 50% confidence level by subjecting two chassis each to the 6000-mile IPT (utilizing the binomial distribution, no durability failures for two vehicles are allowed to meet this requirement).

4.8.3.3 Environmental operation. When specified (see 6.2) conformance to 3.4.3, shall be determined by operating and storing the chassis at specified temperatures. The chassis shall evidence no damage as a result of such operations or storage.

4.8.3.4 Highway operation. To determine conformance to 3.4.4.1 and 3.4.4.2, the chassis shall be operated as specified on paved and gravel roads and observed for towing and performance requirements.

4.8.3.5 Cross-country operation. To determine conformance to 3.4.4.3, the chassis shall be operated over cross-country terrain as specified without damage to the chassis or the prime mover.

4.8.3.6 Grade and slope operation. To determine conformance to 3.4.5, the chassis shall be operated on 40 percent grades headed up and down grade, and on 20 percent side slopes with each side of chassis up-slope.

4.8.3.7 Service brakes. To determine conformance to 3.4.6.1, the vehicle combination shall be operated as specified and observed for distance to stop chassis and towing vehicle. Brake stopping distance shall not exceed that specified.

4.8.3.8 Automatic brake actuation. To determine conformance to 3.4.6.1.1, a fully loaded trailer, coupled to prime mover, shall be parked on a 15% grade. With prime mover brakes applied, disconnect air lines to trailer, allowing actuator to engage trailer brakes. For inertia brake systems, engage trailer brakes by pulling breakaway chain. The prime mover is to be moved to provide clearance between pintle and lunette so that trailer movement is possible. Reset prime mover brakes; trailer must remain stationary for 15 minutes. Any movement prior to 15 minutes shall constitute test failure. Perform this test both uphill and downhill.

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4.8.3.9 Parking brake. To determine conformance to 3.4.6.2, the parking (hand) brake shall be actuated with the chassis on a 30 percent grade, as specified. With service brakes deactivated and trailer coupled to prime mover set trailer parking brakes. Move prime mover to provide clearance between pintle and lunette and set prime mover brakes. No movement at pintle/lunette couple is basis for conformance to requirement for brakes.

4.8.3.10 Landing device and leveling jacks. To determine conformance to 3.4.7 and 3.4.8, the chassis shall be placed on a firm surface and the landing device and leveling mechanisms operated. The towing vehicle shall be coupled and uncoupled and the leveling mechanism raised and lowered.

4.8.3.11 Turning and tracking ability test. To determine conformance to 3.4.9, the prime mover shall be coupled to the chassis and driven to the prime mover's minimum turning circle to the right and to the left, without interference between the prime mover and the towed chassis. During road test, chassis tracking ability shall be observed.

4.8.3.12 Fording ability. To determine conformance to 3.4.10, the chassis shall be towed through water crossings at depth and for the time specified. Chassis and components shall subsequently be examined for water damage.

4.8.3.13 Stabilizers test. To determine conformance to 3.4.11, the chassis, with cross-country payload specified in table II, shall be stabilized as specified on an off-the-road site. Bases shall be extended to the ground to confirm adjustability.

4.8.4 Preservation, packaging and vehicle processing inspection. To determine conformance to Section 5 in this document, material and equipment shall be inspected prior to shipment.

5. Packaging.

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level shall be in accordance with the applicable packaging standard (MIL-V-62038) and packaging data sheet specified by the contracting authority (see 6.2).

6. NOTES

6.1 Intended use. The chassis covered by this specification are intended for mounting of various bodies used by the Armed Services for transporting personnel, supplies and equipment, or as mounts for special equipment during tactical military operations.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Model of chassis required (see 1.2).
c. If first article samples are not required (see 3.1).
d. Title, number, and date of applicalbe drawing (see 3.3).
e. If service brakes are required (3.3.8).
f. If swing-up landing device is not required (3.3.9).

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g. If leveling jacks are required (see 3.4.8).
h. If chassis and component assemblies and parts do not require painting (see 3.5.1).
i. If camouflage pattern is required (see 3.5.1).
j. If rustproofing is not required (see 3.5.2).
k. If responsibility for inspection shall be other than specified (see 4.1).
l. If responsibility for inspection equipment shall be other than specified (see 4.1.2).
m. If special information is required on identification plates (see 3.6.1).
n. Environmental test, when required (4.8.3.3).
o. Applicable levels of preservation, packaging, packing, and marking (see 5.1).

6.3 Recycled materials. The use of recycled materials which meet the requirements of the applicable material specifications without jeopardizing the intended use of the item shall be encouraged (see 3.2).

6.4 Subject term (key word) listing.

Cart, 2-Wheel, Chassis, Trailer, 1/4 to 3 1/2 Ton
Chassis, Trailer, 2-Wheel Cart Type
Trailer, Chassis, 2-Wheel Cart Type
2-Wheel, Cart Type Trailer
1/4 to 3 1/2 Ton Chassis Trailer

6.5 Changes from previous issue. Asterisks are not used in this revision, to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Army - AT
Navy - MC

User activities:
Army - ME
Navy - CG

Preparing activity:
Army - AT

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